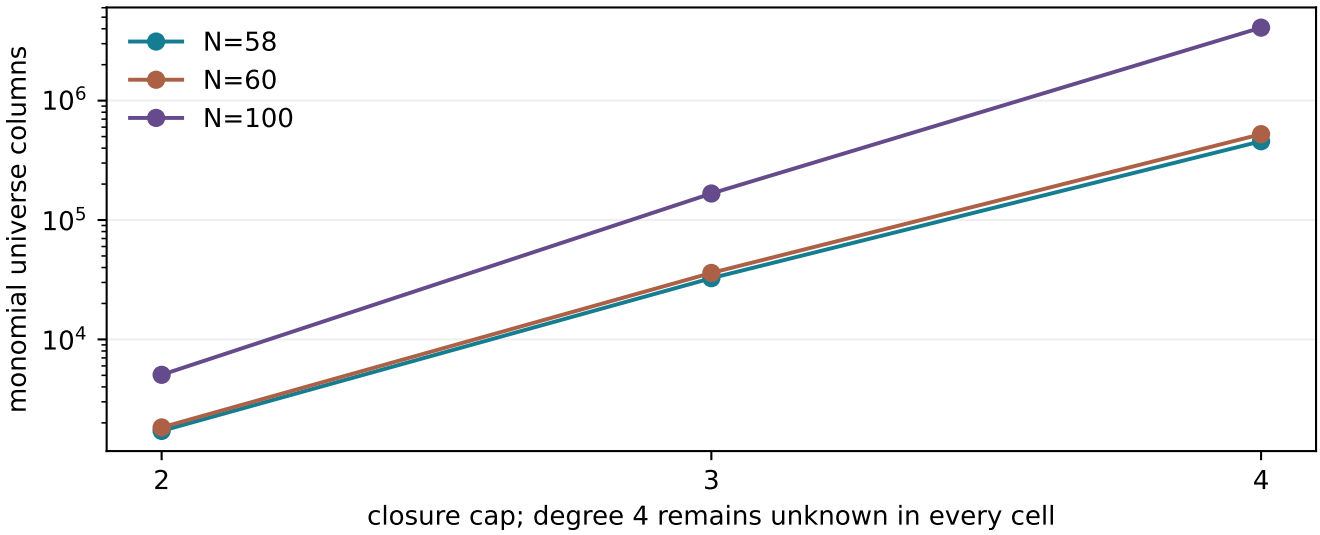
N=58/60/100: fixed points and retained caps

Three registered larger-variable Boolean probes, 2026-10-25. Each uses the frozen first random-family UNSAT draw named by its input manifest: N=58 draw 2, N=60 draw 0, N=100 draw 0. N counts Boolean variables, not field or subgroup bits. Parameters and terminal dimensions:

N	m	field n	ell	dim at D2	dim at D3	D4 result
58	6	10	3	67	11,166	allocation failed
60	5	15	3	74	18,288	allocation failed
100	10	10	2	109	45,344	entry cap



All six degree-two/three closures saturate without refuting. Dense N=58 and N=60 begin degree four and abort with allocation-failure diagnostics under the 12-GiB address-space limit (exit 134). Sparse N=100 begins degree four with an implicit 4,087,976-column universe, then reaches the 25-million-entry cap (process exit 0, scientific verdict incomplete). An ordinary process exit does not establish a fixed point. No degree-four refutation or saturation is claimed for any cell. The sparse universe is implicit, not a dense allocation.

Limits: 900 seconds per command, 12 GiB address space, dense outer cap 600,000 columns; sparse storage has its separate entry cap. Four Rayon threads, HKY_CHUNK=256, shared x86-64 host, L0. The u128/sparse instrument does not remove the production library's 64-variable boundary. Dimensions are native observations; no independent large fixed-point certificate is provided. Exact frozen Boolean solution counts do not establish ordinary curve/subgroup-query relevance, which remains unknown here. No bounded-degree law, complete DLP cost, speedup or hardware result follows.

The native collector checks every input/row hash, declared variable count, process failure and terminal status, and reproduces byte for byte. Source: `examples/ic_hky_probe_summary.rs`. Receipt: `../../pipeline-receipts/hky-probes-20261025/receipt.json`. Raw archive under evidence/: `runs-local-followup-20261025-01.tar.xz`. It contains all three probes and only the separate completed draw-6 pilot; the remaining live branch campaign is not represented as finished. Protocols: PROTOCOL.md and GUESS_CAP200K_PROTOCOL.md. The complete remote continuation and same-input external large-cell checks remain outstanding.